

$$\begin{aligned}
& \frac{1}{16} g_2^2 \left(\frac{1}{12} g_2^2 (3 g_1^2 - 5 g_2^2) + \frac{2}{9} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \right) \text{LF}_{3,0}[\text{m}_2] + \frac{1}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,-1}[\text{m}_2] - \\
& \frac{16}{45} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,-2}[\text{m}_2] + \frac{1}{4} \left(-2 g_2^2 \overline{y_e^{\text{PR}}} y_e^{\text{PR}} + \sum_{\text{p}} g_2^4 \right) \text{LF}_{2,0}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{36} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \left(-18 \overline{y_e^{\text{PR}}} y_e^{\text{PR}} + 13 \sum_{\text{p}} g_2^2 \right) \text{LF}_{3,0}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{24} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \left(12 \overline{y_e^{\text{PR}}} y_e^{\text{PR}} - 11 \sum_{\text{p}} g_2^2 \right) \text{LF}_{4,-1}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{45} \sum_{\text{p}} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,-2}[\text{m}_1^{\text{p}}] + \frac{3}{4} g_2^2 \left(-2 \overline{y_d^{\text{PR}}} y_d^{\text{PR}} - 2 \overline{y_u^{\text{PR}}} y_u^{\text{PR}} + \sum_{\text{p}} g_2^2 \right) \text{LF}_{2,0}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{12} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \left(-18 (\overline{y_d^{\text{PR}}} y_d^{\text{PR}} + \overline{y_u^{\text{PR}}} y_u^{\text{PR}}) + 13 \sum_{\text{p}} g_2^2 \right) \text{LF}_{3,0}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{8} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \left(12 \overline{y_d^{\text{PR}}} y_d^{\text{PR}} + 12 \overline{y_u^{\text{PR}}} y_u^{\text{PR}} - 11 \sum_{\text{p}} g_2^2 \right) \text{LF}_{4,-1}[\text{m}_1^{\text{p}}] + \\
& \frac{1}{45} \sum_{\text{p}} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,-2}[\text{m}_1^{\text{p}}] + \frac{1}{9} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,0}[\tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,-1}[\tilde{\mu}] - \frac{8}{45} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,-2}[\tilde{\mu}] + g_1^4 \tilde{\mu}^2 \text{LF}_{2,-1}[\text{m}_1, \tilde{\mu}] + \\
& \frac{1}{2} \text{m}_1 \tilde{\mu} \text{g}_1^4 (\overline{C_{\phi 102}} + C_{\phi 102}) \text{LF}_{2,2,0}[\text{m}_1, \tilde{\mu}] - \frac{4}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{2,1,0}[\text{m}_2, \tilde{\mu}] - \\
& 2 g_2^4 \text{LF}_{2,-2,-2}[\text{m}_2, \tilde{\mu}] + \frac{1}{3} g_2^4 \left(2 (C_{\phi 1^2} + C_{\phi 2^2} + 3 \text{m}_2^2) + 9 \tilde{\mu}^2 \right) \text{LF}_{2,2,-1}[\text{m}_2, \tilde{\mu}] + \\
& \frac{1}{6} \text{m}_2 g_2^4 \left(-6 \text{m}_2 (C_{\phi 1^2} + C_{\phi 2^2}) + 13 \tilde{\mu} (\overline{C_{\phi 102}} + C_{\phi 102}) \right) \text{LF}_{2,2,0}[\text{m}_2, \tilde{\mu}] + \\
& \frac{2}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,1,-1}[\text{m}_2, \tilde{\mu}] - \frac{10}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,2,-2}[\text{m}_2, \tilde{\mu}] + \\
& \frac{1}{3} \text{m}_2 g_2^4 \left(9 \text{m}_2 (C_{\phi 1^2} + C_{\phi 2^2}) - \tilde{\mu} (\overline{C_{\phi 102}} + C_{\phi 102}) \right) \text{LF}_{3,2,-1}[\text{m}_2, \tilde{\mu}] + \\
& 2 g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,3,-3}[\text{m}_2, \tilde{\mu}] - 2 g_2^4 g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,3,-2}[\text{m}_2, \tilde{\mu}] + \\
& 2 g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,2,-3}[\text{m}_2, \tilde{\mu}] - 2 g_2^4 \text{m}_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,2,-2}[\text{m}_2, \tilde{\mu}] + \\
& 3 \overline{y_d^{\text{PR}}} y_d^{\text{SR}} \overline{y_u^{\text{ST}}} y_u^{\text{PT}} \text{LF}_{1,1,0}[\text{m}_d^{\text{r}}, \text{m}_u^{\text{t}}] - g_2^2 \overline{a_e^{\text{PR}}} a_e^{\text{PR}} \text{LF}_{2,1,0}[\text{m}_1^{\text{p}}, \text{m}_e^{\text{r}}] + \\
& \frac{1}{2} g_2^2 (\overline{a_e^{\text{PR}}} a_e^{\text{PR}} + \tilde{\mu}^2 \overline{y_e^{\text{PR}}} y_e^{\text{PR}}) \text{LF}_{3,1,-1}[\text{m}_1^{\text{p}}, \text{m}_e^{\text{r}}] - \\
& g_2^2 (\overline{C_{\phi 102}} \tilde{\mu} \overline{y_e^{\text{PR}}} a_e^{\text{PR}} + \overline{a_e^{\text{PR}}} (2 C_{\phi 2^2} a_e^{\text{PR}} + C_{\phi 102} \tilde{\mu} y_e^{\text{PR}})) \text{LF}_{3,1,0}[\text{m}_1^{\text{p}}, \text{m}_e^{\text{r}}] + \\
& \frac{1}{4} g_2^2 (\overline{a_e^{\text{PR}}} (18 C_{\phi 2^2} a_e^{\text{PR}} + 13 C_{\phi 102} \tilde{\mu} y_e^{\text{PR}}) + \tilde{\mu} \overline{y_e^{\text{PR}}} (13 \overline{C_{\phi 102}} a_e^{\text{PR}} + 2 \tilde{\mu} y_e^{\text{PR}} (4 C_{\phi 1^2} + C_{\phi 2^2}))) \\
& \text{LF}_{4,1,-1}[\text{m}_1^{\text{p}}, \text{m}_e^{\text{r}}] - \frac{1}{3} g_2^2 (\overline{a_e^{\text{PR}}} (a_e^{\text{PR}} (C_{\phi 1^2} + 7 C_{\phi 2^2}) + 7 C_{\phi 102} \tilde{\mu} y_e^{\text{PR}}) + \\
& \tilde{\mu} \overline{y_e^{\text{PR}}} (7 \overline{C_{\phi 102}} a_e^{\text{PR}} + \tilde{\mu} y_e^{\text{PR}} (7 C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{5,1,-2}[\text{m}_1^{\text{p}}, \text{m}_e^{\text{r}}] - \\
& 3 g_2^2 \overline{a_d^{\text{PR}}} a_d^{\text{PR}} \text{LF}_{2,1,0}[\text{m}_d^{\text{r}}, \text{m}_d^{\text{r}}] + \frac{3}{2} g_2^2 (\overline{a_d^{\text{PR}}} a_d^{\text{PR}} + \tilde{\mu}^2 \overline{y_d^{\text{PR}}} y_d^{\text{PR}}) \text{LF}_{3,1,-1}[\text{m}_d^{\text{r}}, \text{m}_d^{\text{r}}] - \\
& 3 g_2^2 (\overline{C_{\phi 102}} \tilde{\mu} \overline{y_d^{\text{PR}}} a_d^{\text{PR}} + \overline{a_d^{\text{PR}}} (2 C_{\phi 2^2} a_d^{\text{PR}} + C_{\phi 102} \tilde{\mu} y_d^{\text{PR}})) \text{LF}_{3,1,0}[\text{m}_d^{\text{r}}, \text{m}_d^{\text{r}}] + \\
& \frac{3}{4} g_2^2 (\overline{a_d^{\text{PR}}} (18 C_{\phi 2^2} a_d^{\text{PR}} + 13 C_{\phi 102} \tilde{\mu} y_d^{\text{PR}}) + \tilde{\mu} \overline{y_d^{\text{PR}}} (13 \overline{C_{\phi 102}} a_d^{\text{PR}} + 2 \tilde{\mu} y_d^{\text{PR}} (4 C_{\phi 1^2} + C_{\phi 2^2}))) \\
& \text{LF}_{4,1,-1}[\text{m}_d^{\text{r}}, \text{m}_d^{\text{r}}] - \\
& g_2^2 (\overline{a_d^{\text{PR}}} (a_d^{\text{PR}} (C_{\phi 1^2} + 7 C_{\phi 2^2}) + 7 C_{\phi 102} \tilde{\mu} y_d^{\text{PR}}) + \tilde{\mu} \overline{y_d^{\text{PR}}} (7 \overline{C_{\phi 102}} a_d^{\text{PR}} + \tilde{\mu} y_d^{\text{PR}} (7 C_{\phi 1^2} + C_{\phi 2^2}))) \\
& \text{LF}_{5,1,-2}[\text{m}_d^{\text{r}}, \text{m}_d^{\text{r}}] + 3 \overline{y_d^{\text{PR}}} y_d^{\text{SR}} \overline{y_u^{\text{ST}}} y_u^{\text{PT}} \text{LF}_{1,1,0}[\text{m}_d^{\text{p}}, \text{m}_u^{\text{s}}] + \\
& \frac{3}{2} \overline{y_d^{\text{PR}}} y_d^{\text{SR}} \overline{y_u^{\text{ST}}} y_u^{\text{PT}} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{2,1,0}[\text{m}_d^{\text{p}}, \text{m}_u^{\text{s}}] - \frac{3}{2} \overline{y_d^{\text{PR}}} y_d^{\text{SR}} \overline{y_u^{\text{ST}}} y_u^{\text{PT}} (C_{\phi 1^2} + C_{\phi 2^2}) \\
& \text{LF}_{3,1,-1}[\text{m}_d^{\text{p}}, \text{m}_u^{\text{s}}] + \frac{3}{2} g_2^2 (-\overline{a_u^{\text{PR}}} a_u^{\text{PR}} + \tilde{\mu}^2 \overline{y_u^{\text{PR}}} y_u^{\text{PR}}) \text{LF}_{2,1,0}[\text{m}_d^{\text{p}}, \text{m}_u^{\text{r}}] + \\
& \frac{1}{2} g_2^2 (\overline{a_u^{\text{PR}}} (a_u^{\text{PR}} (-2 C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 102} \tilde{\mu} y_u^{\text{PR}}) + \tilde{\mu} \overline{y_u^{\text{PR}}} (\overline{C_{\phi 102}} a_u^{\text{PR}} + \tilde{\mu} y_u^{\text{PR}} (C_{\phi 1^2} + 4 C_{\phi 2^2}))) \\
& \text{LF}_{2,2,0}[\text{m}_d^{\text{p$$